

Chapter 42: On The Structure and Properties of Γ -rings

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Abstract

Let R and Γ be two additive abelian groups. Consider the mapping $R \times \Gamma \times R \rightarrow R$, $(r_1, \gamma, r_2) \mapsto r_1 \gamma r_2$, where $r_1, r_2 \in R$ and $\gamma \in \Gamma$. If this mapping satisfies the following conditions: $(r_1 + r_2) \alpha r_3 = r_1 \alpha r_3 + r_2 \alpha r_3$, $r_1(\alpha + \beta)r_2 = r_1 \alpha r_2 + r_1 \beta r_2$, $r_1 \alpha(r_2 + r_3) = r_1 \alpha r_2 + r_1 \alpha r_3$ and $(r_1 \alpha r_2) \beta r_3 = r_1(\alpha r_2 \beta) r_3 = r_1 \alpha(r_2 \beta r_3)$, then R is called a Γ -ring in the sense of Barnes. Let R and R' be two Γ -ring and $\phi: R \rightarrow R'$ be a mapping. Then ϕ is called a Γ -ring homomorphism if $\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2)$ and $\phi(r_1 \gamma r_2) = \phi(r_1) \gamma \phi(r_2)$ for all $r_1, r_2 \in R$ and $\gamma \in \Gamma$. We prove some important results on Γ -ring homomorphism.

Keywords: Γ -ring, homomorphism, Banach algebras, Γ -ring, Γ -module

1. Introduction

Some significant sets of illuminating objects, such as the set of all rectangular matrices of order $m \times n$, the set of all linear transformations $L(V, V')$ from a vector space V into another vector space V' , etc., seem to lack a natural way to introduce binary algebraic multiplication so that they become an algebra. The concept of a Γ -ring, which is more comprehensive than a ring, was first introduced by N. Nobusawa [5] using the triple product (ternary product), despite the fact that "triple product" attracted independent attention as early as 1949. In an attempt to overcome this challenge, a number of authors, including Barnes [1], Jiang Luh, Lister, Hestenes, G.L. Booth [2], D. K. Bhattacharjya and A. K. Maity, Shoji Kyuno, and N. J. Groenewald [3], developed a ternary system that ultimately led mathematicians to introduce the notations of a Γ -ring, a Γ -module, and a Γ -Banach algebra. Many significant theorems from generic rings, modules, and algebras have been extended to Γ -rings, Γ -modules, and Γ -Banach algebras by a number of eminent mathematicians. We establish several significant findings regarding Γ -ring homomorphism.

2. Preliminaries

Definition 2.1: [5] Let R and Γ be two additive abelian groups. Consider the mapping $R \times \Gamma \times R \rightarrow R$, $(r_1, \gamma, r_2) \mapsto r_1 \gamma r_2$, where $r_1, r_2 \in R$ and $\gamma \in \Gamma$. If this mapping satisfies the following conditions:

$$N_1. (r_1 + r_2) \alpha r_3 = r_1 \alpha r_3 + r_2 \alpha r_3, r_1(\alpha + \beta)r_2 = r_1 \alpha r_2 + r_1 \beta r_2, r_1 \alpha(r_2 + r_3) = r_1 \alpha r_2 + r_1 \alpha r_3,$$

$$N_2. (r_1 \alpha r_2) \beta r_3 = r_1(\alpha r_2 \beta) r_3 = r_1 \alpha(r_2 \beta r_3)$$

and $N_3.$ if $r_1 \alpha r_2 = 0$ for all $r_1, r_2 \in R$, then $\alpha = 0$.

Then R is called a Γ -ring. This definition was given by N. Nobusawa in 1964.

In 1966, W. E. Barnes define Γ – ring as follows:

[1] Let R and Γ be additive abelian groups. Consider the mapping $R \times \Gamma \times R \rightarrow R$, $(r_1, \gamma, r_2) \mapsto r_1 \gamma r_2$, where $r_1, r_2 \in R$ and $\gamma \in \Gamma$. If this mapping satisfies the following conditions:

$$B_1. (r_1 + r_2) \alpha r_3 = r_1 \alpha r_3 + r_2 \alpha r_3, r_1 (\alpha + \beta) r_2 = r_1 \alpha r_2 + r_1 \beta r_2, r_1 \alpha (r_2 + r_3) = r_1 \alpha r_2 + r_1 \alpha r_3 \text{ and}$$

$B_2. (r_1 \alpha r_2) \beta r_3 = r_1 (\alpha r_2 \beta) r_3 = r_1 \alpha (r_2 \beta r_3)$, then R is called a Γ -ring in the sense of Barnes. This definition is slightly weaker than the origin one due to Nobusawa [7].

The operation $r_1 \gamma r_2$ is called Γ -multiplication. Every ordinary ring is a Γ -ring by taking $\Gamma = \mathbb{Z}$ and defining $r_1 n r_2 = n(r_1 r_2)$ for all $n \in \mathbb{Z}$.

Definition 2.2 [2] A nonempty subset $S \subseteq R$ is a Γ -subring if it is closed under subtraction and Γ -multiplication.

Definition 2.3 [3] A subgroup I of R is a Γ -ideal if $R\Gamma I \subseteq I$ and $I\Gamma R \subseteq I$.

Definition 2.4 [2] Let R and R' be two Γ -ring and $\phi: R \rightarrow R'$ be a mapping. Then ϕ is called a Γ -ring homomorphism if $\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2)$ and $\phi(r_1 \gamma r_2) = \phi(r_1) \gamma \phi(r_2)$ for all $r_1, r_2 \in R$ and $\gamma \in \Gamma$.

Definition 2.5 [2] Let R and R' be two Γ -ring and $\phi: R \rightarrow R'$ be a mapping. Then ϕ is called a Γ -ring isomorphism if (1) ϕ is Γ -ring homomorphism, (2) ϕ is injective and (3) ϕ is surjective. If ϕ is an isomorphism from Γ -ring R into a Γ -ring R' , then we called R is isomorphic to R' and written as $R \cong R'$. If ϕ is Γ -ring homomorphism from R onto R' , then ϕ is called an epimorphism and R' is isomorphic image of R . The set $\{x \in R: \phi(x) = 0\}$ is called kernel of ϕ and it is denoted by $\ker \phi$. $\ker \phi$ is a Γ -ideal of R and $Im(\phi) = \{\phi(x): x \in R\}$ is a Γ -subring of R' .

Definition 2.6 Let I be a Γ -ideal of R . Define the equivalence relation $r_1 \sim r_2$ iff $r_1 - r_2 \in I$. The equivalence class of r is $r + I = \{r + i: i \in I\}$. The quotient set $R/I = \{r + I: r \in R\}$ becomes a Γ -ring under the operations $(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I$ and $(r_1 + I) \gamma (r_2 + I) = (r_1 \gamma r_2) + I$ for all $r_1, r_2 \in R$ and $\gamma \in \Gamma$.

3. Main Results

Theorem 3.1 Let $\phi: R \rightarrow R'$ be a Γ -ring homomorphism and let $K = \ker \phi$. Then $R/K \cong Im(\phi)$.

Proof: Define $\psi: \frac{R}{K} \rightarrow Im(\phi)$ by $(r + K) = \phi(r)$.

ψ is Well-defined: Let $r_1 + K = r_2 + K$. Then

$$r_1 - r_2 \in K$$

$$\Rightarrow \phi(r_1 - r_2) = 0$$

$$\Rightarrow \phi(r_1) = \phi(r_2) \text{ as } \phi \text{ be a } \Gamma\text{-ring homomorphism.}$$

$$\Rightarrow \psi(r_1 + K) = \psi(r_2 + K).$$

Therefore ψ is well-defined.

ψ is Γ -ring homomorphism: Let $r_1 + K, r_2 + K \in R/K$ and $\gamma \in \Gamma$. Then

$$\begin{aligned} \psi((r_1 + K) + (r_2 + K)) &= \psi((r_1 + r_2) + K) \\ &= \phi(r_1 + r_2) \\ &= \phi(r_1) + \phi(r_2) \text{ as } \phi \text{ be a } \Gamma\text{-ring homomorphism.} \\ &= \psi(r_1 + K) + \psi(r_2 + K) \end{aligned}$$

$$\begin{aligned} \text{Also } \psi((r_1 + K)\gamma(r_2 + K)) &= \psi((r_1\gamma r_2) + K) \\ &= \phi(r_1\gamma r_2) \\ &= \phi(r_1)\gamma\phi(r_2) \text{ as } \phi \text{ be a } \Gamma\text{-ring homomorphism.} \\ &= \psi(r_1 + K)\gamma\psi(r_2 + K) \end{aligned}$$

ψ is injective: If $\psi(r + K) = 0$, then $\phi(r) = 0$, so $r \in K$. Hence $r + K = K$, the zero element of R/K . Thus $\ker(\psi) = \{K\}$ and ψ is injective.

ψ is surjective: Every element of $Im(\phi)$ has the form $\phi(r) = \psi(r + K)$. Hence ψ is surjective.

Since ψ is a bijective Γ -ring homomorphism, $R/K \cong Im(\phi)$.

Theorem 3.2 If $f: R \rightarrow R'$ is a Γ -ring onto homomorphism with $\ker f = K$, S' a Γ -subring of R' and $S = \{r \in R: f(r) \in S'\} = f^{-1}(S')$, then S is a Γ -subring of R containing K and $R/S \cong R'/S'$.

Proof: First we prove S is a Γ -subring of R containing K . Let $s_1, s_2 \in S$. Since S' a Γ -subring of R' , therefore

$$\begin{aligned} f(s_1), f(s_2) &\in S' \\ \Rightarrow f(s_1) - f(s_2) &\in S' \text{ as } S' \text{ a } \Gamma\text{-subring of } R'. \\ \Rightarrow f(s_1 - s_2) &\in S' \text{ as } f \text{ is a } \Gamma\text{-ring homomorphism.} \\ \Rightarrow s_1 - s_2 &\in S \end{aligned}$$

Again, let $\gamma \in \Gamma$. Since S' a Γ -subring of R' , therefore

$$f(s_1)\gamma f(s_2) \in S'$$

$\Rightarrow f(s_1 \gamma s_2) \in S'$ as f is a Γ -ring homomorphism.

$$\Rightarrow s_1 \gamma s_2 \in S$$

Therefore S is a Γ -subring of R . Let $r \in K = \ker f$. Then $f(r) = 0_{R'} \in S'$ as S' a Γ -subring of R' .

Which implies $r \in S$. Hence $K \subseteq S$.

Define $\psi: R \rightarrow R'/S'$ by $\psi(r) = f(r) + S'$.

ψ is Well-defined: Let $r_1 = r_2$. Then $f(r_1) = f(r_2)$ and which implies

$$\Rightarrow f(r_1) + S' = f(r_2) + S'$$

$$\Rightarrow \psi(r_1) = \psi(r_2).$$

Therefore ψ is well-defined.

ψ is Γ -ring homomorphism: Let $r_1, r_2 \in R$ and $\gamma \in \Gamma$. Then

$$\begin{aligned} \psi(r_1 + r_2) &= f(r_1 + r_2) + S' \\ &= [f(r_1) + f(r_2)] + S' \\ &= [f(r_1) + S'] + [f(r_2) + S'] \\ &= \psi(r_1) + \psi(r_2) \end{aligned}$$

Also

$$\begin{aligned} \psi(r_1 \gamma r_2) &= f(r_1 \gamma r_2) + S' \\ &= [f(r_1) \gamma f(r_2)] + S' \\ &= [f(r_1) + S'] \gamma [f(r_2) + S'] \\ &= \psi(r_1) \gamma \psi(r_2) \end{aligned}$$

Hence ψ is a Γ -ring homomorphism.

ψ is surjective: Let $r' + S' \in R'/S'$. Since $f: R \rightarrow R'$ is onto, therefore there exists $r \in R$ such that $f(r) = r'$ and hence $r' + S' = f(r) + S' = \psi(r)$. Thus ψ is surjective.

Thus ψ is an epimorphism and hence by Theorem 3.1, $R/\ker \psi \cong R'/S'$, where

$$\begin{aligned} \ker \psi &= \{ r \in R: \psi(r) = S' \} \\ &= \{ r \in R: f(r) + S' = S' \} \\ &= \{ r \in R: f(r) \in S' \} \\ &= \{ r \in R: r \in S \} \\ &= S \end{aligned}$$

Lemma 3.3 If S is a Γ -subring of R and T a Γ -ideal of a Γ -ring R , then $S \cap T$ is a Γ -ideal of S and $S + T$ is a Γ -subring of R .

Theorem 3.4 If S is a Γ -subring of R and T a Γ -ideal of a Γ -ring R , then $(S \cap T) \cong (S + T)/T$.

Proof: Define $\phi: S \rightarrow (S + T)/T$ by $\phi(s) = s + T$.

ϕ is Well-defined: Since $s \in S \Rightarrow s \in S + T$, the coset $s + T$ is in $(S + T)/T$.

ϕ is Γ -ring Homomorphism: Let $s_1, s_2 \in S$ and $\gamma \in \Gamma$. Then

$$\phi(s_1 + s_2) = (s_1 + s_2) + T = (s_1 + T) + (s_2 + T) = \phi(s_1) + \phi(s_2)$$

$$\text{Also } \phi(s_1 \gamma s_2) = (s_1 \gamma s_2) + T = (s_1 + T) \gamma (s_2 + T) = \phi(s_1) \gamma \phi(s_2)$$

ϕ is surjective: Every element of $(S + T)/T$ has the form $(s + t) + T = s + T = \phi(s)$. Hence ϕ is surjective.

Therefore ϕ is an epimorphism and hence by Theorem 3.1, $(S \cap T)/T \cong (S + T)/T$.

Where $\ker \phi = \{s \in S: \phi(s) = T\} = \{s \in S: s + T = T\} = \{s \in S: s \in T\} = S \cap T$.

Lemma 3.5 If $I \subseteq J$ and J is a Γ -ideal of Γ -ring R , then J/I is a Γ -ideal of Γ -ring R/I .

Theorem 3.6 If I and J are Γ -ideals of a Γ -ring R with $I \subseteq J$, then $(\frac{R}{I}) / (\frac{J}{I}) \cong R/J$.

Proof: Define $\psi: R/I \rightarrow R/J$ by $\psi(r + I) = r + J$.

ψ is Well-defined: Let $r + I = s + I \Rightarrow r - s \in I \subseteq J \Rightarrow r + J = s + J \Rightarrow \psi(r + I) = \psi(s + I)$.

Therefore ψ is well-defined.

ψ is Γ -ring Homomorphism: Let $r + I, s + I \in R/I$ and $\gamma \in \Gamma$. Then

$$\begin{aligned} \psi((r + I) + (s + I)) &= \psi((r + s) + I) \\ &= (r + s) + J \\ &= (r + J) + (s + J) \\ &= \psi(r + I) + \psi(s + I) \end{aligned}$$

Also

$$\psi((r + I) \gamma (s + I)) = \psi((r \gamma s) + I) = (r \gamma s) + J = (r + J) \gamma (s + J) = \psi(r + I) \gamma \psi(s + I)$$

Thus ψ is Γ -ring Homomorphism.

ψ is surjective: For every $r + J \in R/J$, there exists $r + I \in R/I$ such that $\psi(r + I) = r + J$. Hence ψ is surjective.

Thus ψ is an epimorphism and hence by Theorem 3.1, $(R/I)/\ker \psi \cong R/J$, where

$$\begin{aligned} \ker \psi &= \left\{ r + I \in \frac{R}{I} : \psi(r + I) = J \right\} \\ &= \left\{ r + I \in \frac{R}{I} : r + J = J \right\} \end{aligned}$$

$$= \left\{ r + I \in \frac{R}{I} : r \in J \right\}$$

$$= \frac{J}{I}$$

4. Conclusion

The conclusions given above are predicated on a fundamental knowledge of rings, Γ -rings, and additive abelian groups. We prove several significant theorems on Γ -ring in this work. This study has broadened our understanding in a more thorough way, and we have found it to be really exciting. Perhaps it will inspire others to carry out additional research in this field. Additionally, the author believes that by using alternative methods, the results could be expanded in the future.

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